A Context-Aware Self-Healing Security Framework for WBAN Against Sybil Attacks

Chanuka B.D.K.
dept. Computer Engineering
University of Peradeniya
Peradeniya, Sri Lanka
e20049@eng.pdn.ac.lk

Ranawaka R.A.D.J.I.
dept. Computer Engineering
University of Peradeniya
Peradeniya, Sri Lanka
e20318@eng.pdn.ac.lk

Sewmini K.T.R.
dept. Computer Engineering
University of Peradeniya
Peradeniya, Sri Lanka
e20370@eng.pdn.ac.lk

Karunarathna G.S.N.
dept. Computer Engineering
University of Peradeniya
Peradeniya, Sri Lanka
namal@eng.pdn.ac.lk

***Abstract*—Wireless Body Area Networks (WBANs) are central to the Internet of Medical Things (IoMT), enabling continuous physiological monitoring and real-time clinical decision-making. However, the resource-constrained nature of WBAN devices and reliance on wireless communication make them highly vulnerable to identity-based threats, particularly Sybil attacks. By generating multiple fake identities, a Sybil node can distort routing paths, disrupt trust mechanisms, inject falsified medical data, and compromise patient safety. This review analyzes existing Sybil attack detection and mitigation approaches within IoMT-driven systems, categorizing solutions into trust-based, context-aware, cryptographic, and machine learning approaches. While recent research demonstrates improved detection capabilities, the findings reveal critical limitations, including high computational overhead, static decision models, limited contextual reasoning, and the absence of autonomous recovery mechanisms. Furthermore, most existing solutions operate as detection-only systems rather than resilient, self-healing security frameworks. The review concludes by identifying key research gaps and motivates the need for a lightweight, context-aware, AI-driven self-healing architecture tailored for the operational and clinical constraints of medical WBAN environments.**

***Index Terms*—Wireless Body Area Network (WBAN); Internet of Medical Things (IoMT); Sybil Attack; Context-Aware Security; Lightweight Cryptography; Machine Learning; Trust Management; Self-Healing Security; Medical Sensor Networks; Identity Spoofing; Remote Healthcare Systems.**

I. Introduction

The integration of miniature sensor technology directly into or onto the human body has established the Wireless Body Area Network (WBAN) as a key technology in modern healthcare. WBANs form a foundational layer within the broader Internet of Medical Things (IoMT) ecosystem. WBANs enable real-time health monitoring, remote diagnostics, and immediate intervention for life-critical conditions,fundamentally transforming healthcare from reactive treatment to proactive wellness management. As WBAN-based IoMT systems become more deeply integrated into clinical workflows, ensuring their security becomes essential—not only to protect sensitive patient information, but also to maintain trust in automated medical decision-making processes. However, this shift toward increased interconnectivity and the wireless transmission of sensitive physiological data has introduced significant cybersecurity risks.

The specialized characteristics of WBANs compared to conventional wireless networks include severe energy constraints of implanted devices, limited computational power, dynamic, high-mobility topologies, and the life-critical nature of the data. These characteristics make traditional network security mechanisms, such as heavy cryptographic protocols or centralized detection systems, unsuitable or ineffective. Among the most critical and difficult-to-detect threats exploiting these weaknesses is the Sybil attack, where a single malicious node forges multiple identities to corrupt routing paths, inject false data, or trigger resource depletion, directly compromising the confidentiality, integrity, and availability of patient data.

This literature review synthesizes the current research landscape to identify foundational principles and technical deficiencies in existing WBAN and IoMT security models. It is structured to systematically build a compelling justification for novel work. The following thematic sections will cover:

- WBAN and IoMT Architecture and Constraints: Detailing the layers and unique resource limitations that define the operating environment.
- Security Challenges and Sybil Attack Taxonomy: Classifying the unique threats, particularly the behavior and mechanism of Sybil and spoofing attacks in resource-constrained networks.
- Context-Aware Detection and Feature Engineering: Analyzing methods that utilize both network traffic and physiological sensor data to intelligently distinguish between a genuine medical emergency, a sensor fault, and a malicious intrusion.
- Adaptive Security and Self-Healing Mechanisms: Reviewing contemporary efforts in designing responsive systems and highlighting the limitations of current detection-only models.

By critically analyzing these themes, this review will articulate the specific gaps in the literature concerning a truly lightweight, context-aware, self-healing security framework capable of autonomously detecting and mitigating Sybil attacks while respecting the strict power and computational constraints of WBAN nodes. This synthesis will serve as the technical and academic justification for the proposed research project.

To fully understand the threat landscape and current mitigation strategies, it is essential to first examine the fundamental architecture, constraints, and operational characteristics of IoMT and WBAN systems.

II. Background

This section provides an integrated overview of the key concepts, technologies, and research trends underpinning this study. It introduces the Internet of Medical Things (IoMT) and Wireless Body Area Networks (WBANs), outlines their architectural and operational characteristics, and examines the security challenges that emerge in such resource- and energy-constrained medical environments. It then reviews existing Sybil attack defense mechanisms across trust-based, cryptographic, context-aware, and machine learning–based approaches, highlighting their strengths and limitations. Together, these sections motivate the need for a lightweight, context-aware, and self-healing security framework tailored to WBAN-centric IoMT systems. We begin by exploring the Internet of Medical Things, the overarching ecosystem within which WBANs operate, detailing its layers, functions, and associated security considerations.

A. Internet of Medical Things (IoMT)

The Internet of Medical Things (IoMT) denotes the convergence of medical devices with Internet of Things (IoT) technologies to form interconnected healthcare ecosystems capable of continuous monitoring, remote diagnosis, and real-time clinical decision support. IoMT environments integrate wearable sensors, implantable devices, smart monitoring equipment, and healthcare applications through internet-enabled communication channels, enabling automated collection of physiological data such as ECG, glucose levels, oxygen saturation, and blood pressure [1], [2]. Demand for IoMT has grown rapidly due to its ability to deliver faster, lower-cost, and more personalized healthcare services, a role that became particularly visible during the COVID-19 pandemic, when telehealth and remote patient management were essential to sustaining clinical operations with minimal physical contact [1]. IoMT functionality is typically organized into a multilayer architecture comprising:

- Things Layer – Patient-side devices such as sensors, wearables, actuators, and medical records systems responsible for capturing raw physiological information.
- Fog Layer – Edge or near-patient computing resources that support local processing, low-latency decision-making, preliminary analytics, and first-level security handling, thereby reducing delay and offloading work from constrained devices.
- Cloud Layer – High-capacity infrastructure providing large-scale storage, advanced analytics, predictive modeling, and long-term electronic health record management. This layer enables sophisticated diagnostic support, workflow automation, and clinical decision-support mechanisms [1].

Together, these layers establish a continuous data flow between patients and healthcare providers, supporting applications such as smart hospitals, telemedicine, remote robot-assisted surgery, and personalized treatment systems [1], [2]. The extensive deployment of IoMT devices—ECG monitors, smart wearables, insulin pumps, pulse oximeters, and similar systems—has transformed remote healthcare by allowing clinicians to access real-time data independent of the patient's physical location or device type [2]. This persistent connectivity improves early detection, emergency response, and long-term disease management; for instance, IoMT systems can automatically notify clinicians or trigger emergency services when abnormal vital patterns are detected [2].

At the same time, this pervasive connectivity introduces significant security and privacy risks. IoMT devices typically rely on wireless technologies such as Bluetooth, Wi-Fi, NFC, RFID, LTE, and emerging 5G/6G networks. If strong encryption and authentication are absent or improperly implemented, adversaries can intercept, manipulate, or replay medical information [1], [2]. Mushtaq, Shah, and Ghafoor demonstrate that attackers may forge emergency signals, inject falsified health data, spoof device identities, or disrupt communication channels, leading to misdiagnosis or dangerous delays in treatment [2]. These risks are especially acute for patients who depend entirely on remote monitoring and automated alerts.

The global expansion of IoMT—driven by aging populations, the proliferation of smart healthcare infrastructure, and widespread telemedicine adoption—has positioned it as a rapidly growing, multi-billion-dollar sector [2]. However, this growth magnifies challenges related to large-scale data management, interoperability among heterogeneous devices, scalability, and the protection of highly sensitive health information. Patient privacy remains a central concern, particularly for conditions such as chronic illnesses, infectious diseases, or psychiatric disorders, where confidentiality breaches may carry severe social, ethical, and legal consequences [1], [2].

IoMT devices are also constrained by limited memory, low processing power, and dependence on small batteries. These hardware limitations make conventional cryptographic schemes (e.g., RSA) difficult to deploy without overburdening devices [2]. Consequently, IoMT ecosystems are exposed to a wide range of threats, including malware and botnets (e.g., Mirai), sensor spoofing, replay attacks, unauthorized access, and ransomware [2]. These attacks jeopardize confidentiality, integrity, and availability, underscoring the need for specialized, lightweight, and context-sensitive security mechanisms that match the constraints of medical IoT environments.

Within this broader IoMT ecosystem, Wireless Body Area Networks form a critical sub-layer, responsible for direct physiological data acquisition and real-time patient monitoring.

B. Wireless Body Area Networks (WBAN)

Wireless Body Area Networks (WBANs) are low-power, short-range sensor networks deployed on, in, or around the human body to support continuous monitoring of physiological signals and health status. WBANs form a crucial subset of

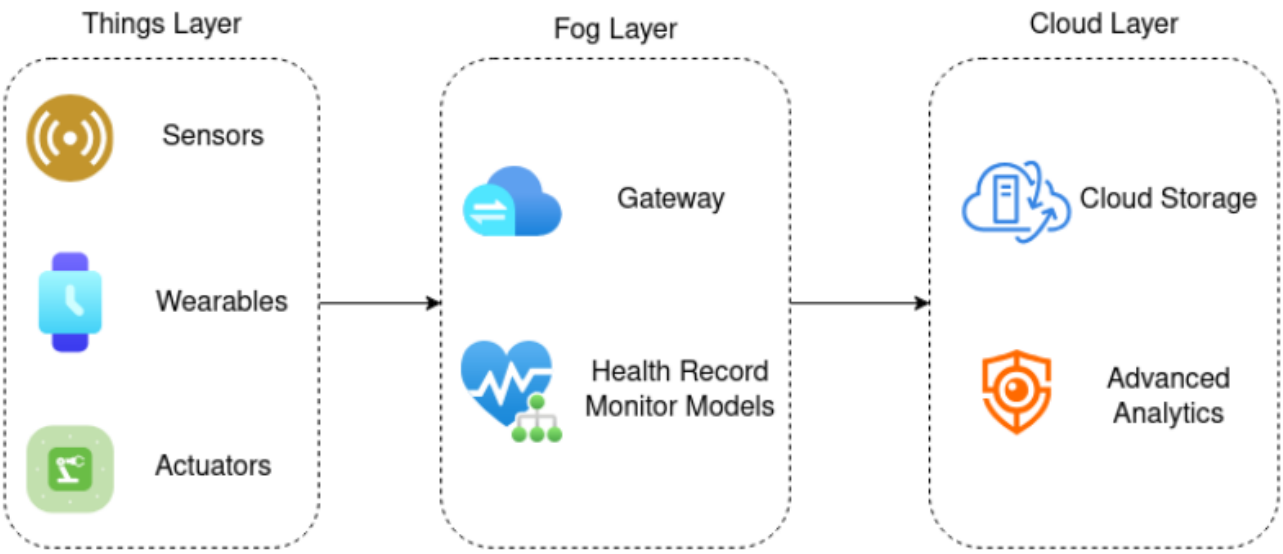

Fig. 1. Multilayer architecture of IOMT system.

IoMT, enabling remote diagnostics, telemedicine, and real-time patient monitoring.

Most WBAN architectures follow a three-tier model:

- Tier 1 – Intra-body communication: Sensors attached to or implanted in the body communicate with each other and with a central coordinator (e.g., a hub node).
- Tier 2 – Personal gateway: The coordinator exchanges data with a gateway device, commonly a smartphone, smartwatch, or specialized hub.
- Tier 3 – External network / cloud: The gateway forwards data to hospital servers, cloud platforms,or healthcare provider systems for storage, analysis, and decision support.

Alternative models—including body-to-body (B2B), machine-to-machine (M2M), and ad-hoc networking—have been explored to extend coverage and support mobility. Within WBANs, various network topologies (star, mesh, tree, hybrid) are used, each presenting different trade-offs in terms of scalability, latency, robustness, and fault tolerance. Interoperability across devices and layers is supported through standards such as IEEE 802.15.1 and IEEE 802.15.4, with IEEE 802.15.6 specifically targeting WBAN communication.

Research shows that improving WBAN performance requires careful handling of mobility, interference, energy consumption, and the limited resources of miniaturized sensors. Proposed techniques include dynamic channel allocation, traffic-aware scheduling, adaptive MAC protocols, and the use of Software-Defined Networking (SDN) to manage traffic and routing. Despite these advances, WBANs remain highly susceptible to communication disruptions and security threats, due to their proximity-based communication model, stringent energy budgets, and the critical nature of medical data.

Security analyses identify a variety of attacks targeting WBANs across the physical, MAC, and network layers, including eavesdropping, jamming, and particularly Sybil attacks, where one malicious device forges multiple identities to corrupt routing, disrupt trust mechanisms, and manipulate network behavior. Because WBAN data are time-sensitive and often life-critical, faults or attacks can directly impact clinical decisions. Recent surveys conclude that traditional fault-tolerance and security approaches from generic wireless sensor networks are not directly applicable to WBANs due to their extreme energy constraints, continuous mobility, and strict latency requirements. These constraints motivate the development of lightweight, context-aware, and self-healing security mechanisms specifically designed for medical body-area communication.

While WBANs enable life-critical health monitoring, their resource constraints and wireless nature introduce unique security challenges that must be addressed to maintain data integrity and patient safety.

The WBAN 3-Tier Architecture is shown in Figure 2.

C. Security Challenges in IoMT/WBAN

The expansion of continuous, wireless physiological monitoring in IoMT and WBAN systems has significantly broadened both the capabilities of remote healthcare and the exposure to security risks [3], [4]. These networks operate over short-range wireless channels [23] and depend on devices with severe limitations in computation, memory, and battery capacity [5], resulting in a more fragile security posture compared to traditional clinical IT infrastructures.

Confidentiality, integrity, and availability remain the core security requirements for these systems [3], yet achieving them is challenging due to several structural constraints. First, IoMT deployments are heterogeneous: sensors from different vendors, using diverse communication standards, must interoperate securely [6]. Second, short-range wireless links such as

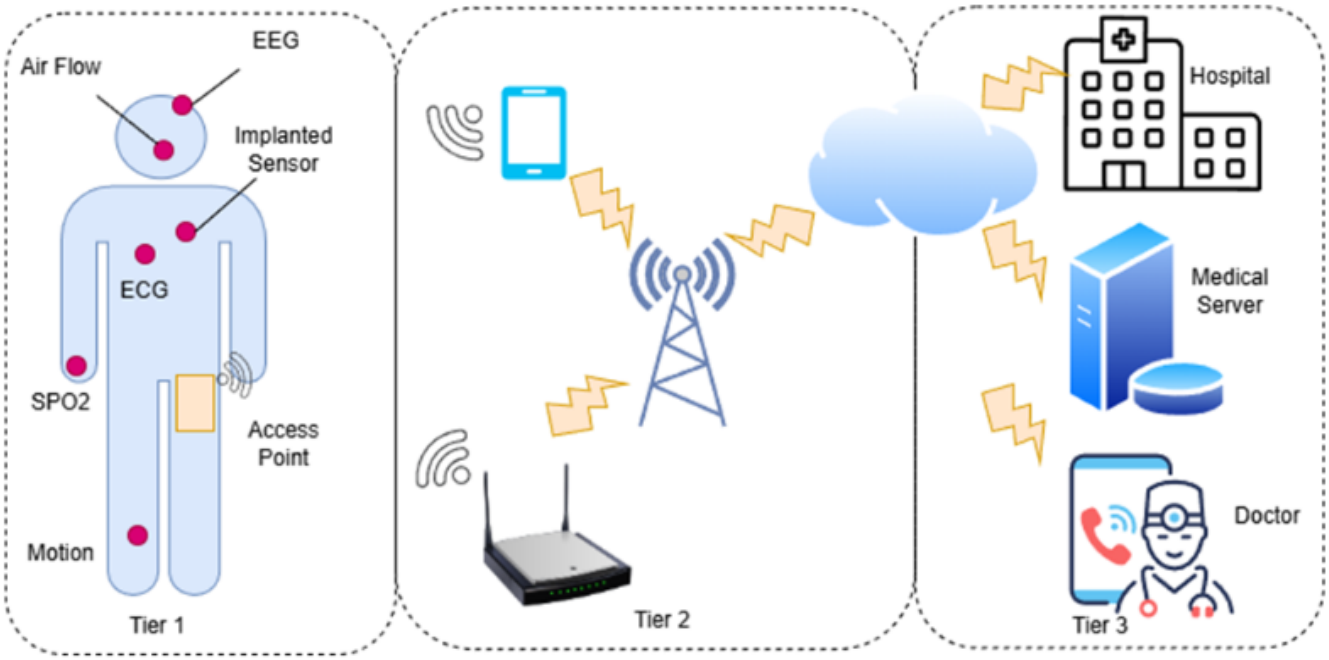

Fig. 2. WBAN 3-Tier Architecture.

Bluetooth and IEEE 802.15.6 are vulnerable to eavesdropping, inference attacks, and jamming [5], [7], [8]. Finally, many WBAN routing protocols lack robust identity verification, making them susceptible to spoofing, selective forwarding, and false data injection attacks [5], [9].

Among the various threats, Sybil attacks are particularly disruptive [4]. By forging multiple identities [10], a malicious node can manipulate routing decisions, distort network topology [11], and artificially inflate trust scores [9]. This directly compromises data reliability and may endanger clinical decision-making when corrupted physiological readings are incorporated into medical workflows. Other risks such as tampering, replay attacks, and denial-of-service become even more serious due to the lack of standardized baseline security practices across device manufacturers [12].

In response, recent research has explored lightweight cryptography optimized for constrained hardware [3], [4], [6], [10], anomaly- and behavior-based intrusion detection [6], [13], and decentralized trust management [15], including blockchain-supported identity validation and AI-enabled intrusion detection systems [13]. However, each of these approaches introduces trade-offs: blockchain methods impose communication and synchronization overhead [4], [14]; machine-learning-based IDS models require substantial training data and may be computationally intensive [12], [14]; and cryptographic enhancements risk shortening device lifetime [6]. Balancing security, energy efficiency, latency, and privacy therefore remains a central design challenge in IoMT/WBAN security research [4], [5], [15]–[17].

Given these limitations, Sybil attacks require focused attention, as their ability to compromise identity and routing [10] makes them uniquely harmful in resource-constrained medical sensing environments [15].

D. Sybil Attacks in Medical WBAN

The Sybil attack is a major identity-based threat in WBAN and IoMT networks, where a single malicious device forges multiple false identities to appear as numerous legitimate nodes. By injecting these fabricated identities into routing, clustering, or trust-management systems, the attacker gains disproportionate control over network operations, enabling route manipulation, false data injection, and large-scale disruption with minimal resources [11]. Because WBAN/IoMT consists of small, resource-constrained nodes carrying life-critical physiological data, even a small number of Sybil identities can destabilize routing or distort medical readings [15].

E. Sybil attack variants in IoMT/WBAN

Recent studies categorize Sybil attacks into three models as SA-I, SA-II, and, SA-III based on attacker distribution and mobility:

- SA-I (4): Restricted / Localized Static Sybil Attack
  - A single attacker influences a small physical region, introducing Sybil identities that interact primarily with nearby sensors [11].
  - Relevance: Highly applicable to WBAN, which operates within a small physical radius around the human body [10].
- SA-II (5): Distributed Static Sybil Attack

- Multiple static attacker nodes are dispersed across the network, each generating Sybil identities and creating multiple attack edges [11].
- Relevance: Particularly threatening in IoMT systems involving multiple wearables and gateways, enabling large-scale trust degradation and route manipulation [11].

- SA-III (6): Distributed Mobile Sybil Attack
  - Attacker nodes are mobile and inject Sybil identities while moving. The result is continuously shifting topology manipulation.
  - Relevance: Highly impactful in mobile-health IoMT settings (patients walking or moving within hospital spaces), where node mobility already complicates routing [11].

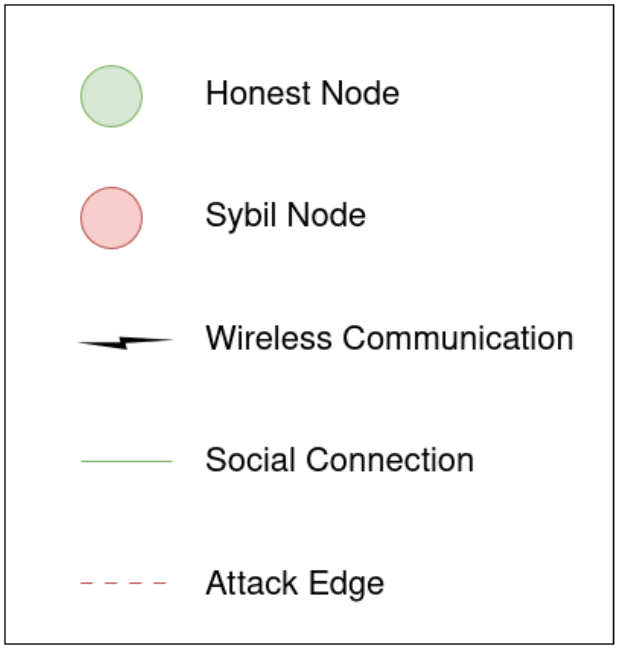

Fig. 3. Legend.

Together, these models show that Sybil attacks can be localized or distributed, static or mobile, with SA-II and SA-III posing the highest risk in IoMT/WBAN due to their ability to affect larger portions of the network and exploit mobility patterns.

F. Real-World Impact of Sybil Attacks on Medical WBAN

Because medical WBANs rely on continuous physiological monitoring (ECG, heart rate, temperature, etc.) [18], Sybil attacks directly threaten the reliability and timeliness of critical patient data. [19] demonstrates that introducing even 10% Sybil nodes significantly degrades network performance in commonly used WBAN routing protocols such as AODV and OLSR:

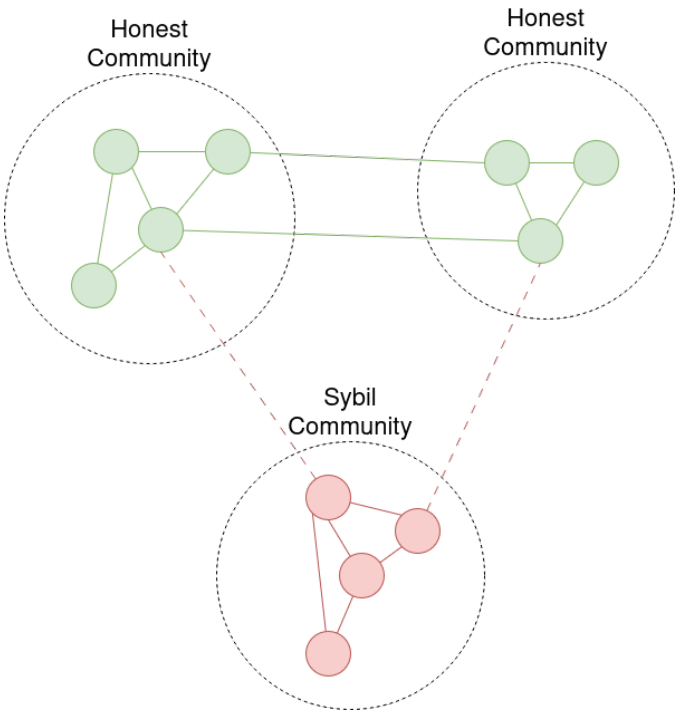

Fig. 4. SA-I.

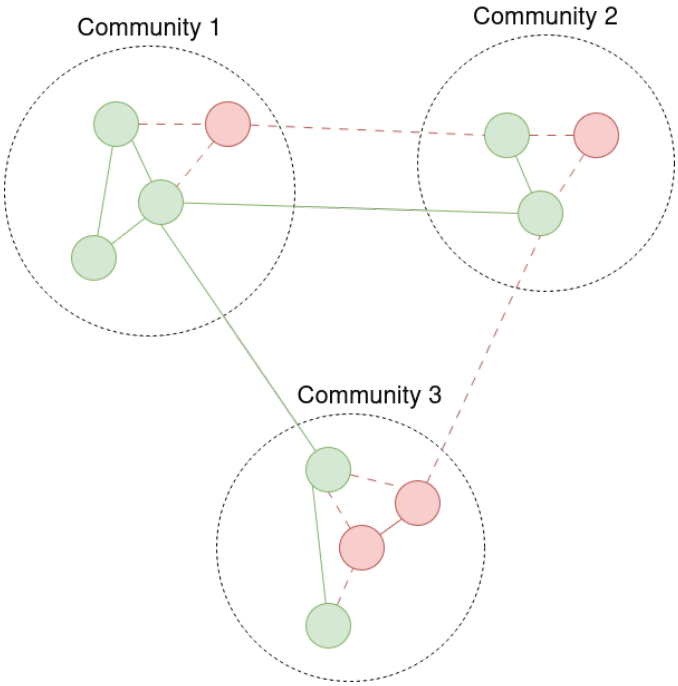

Fig. 5. SA-II.

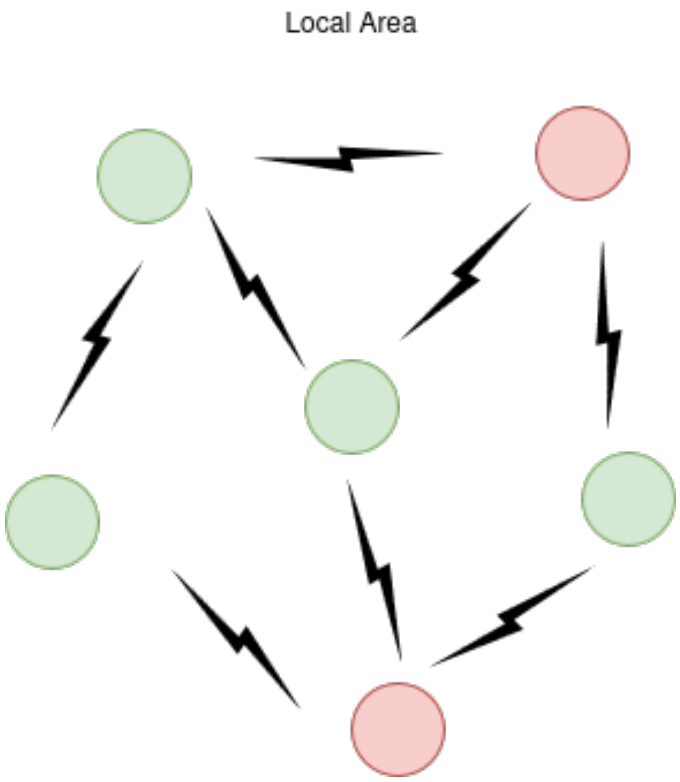

Fig. 6. SA-III.

- Packet Delivery Ratio decreases (AODV drops from 98% $\rightarrow$ 72%), risking missed or delayed transmission of health data [4], [20].
- End-to-end delay increases (+34% for AODV), which is critical for real-time monitoring. [20]
- Energy consumption rises, reducing wearable device lifetime [15].
- Routing overhead increases, as nodes waste resources maintaining connectivity under identity fraud [11].

These impacts demonstrate that Sybil attacks compromise WBAN reliability, increase latency, and reduce energy efficiency—all of which may delay alerts, mislead automated decision systems, or cause incorrect patient monitoring [**?**]. This highlights the need for lightweight, security-aware detection and mitigation tailored to medical sensor constraints [6], [11], [15], [18].

III. Existing Defense Techniques Against Sybil Attacks

Given these severe consequences, researchers have proposed multiple defense mechanisms in IoMT/WBAN environments can be broadly grouped into four categories: trust-based, context-aware, cryptographic/blockchain-based, and machine learning–based approaches. Each category addresses different aspects of the problem and exhibits distinct trade-offs in accuracy, overhead, and suitability for constrained medical devices.

A. Trust-Based Approaches

Trust-based mechanisms detect Sybil behavior by monitoring node actions and assigning trust scores based on behavioral evidence rather than static credentials. These models are attractive in dynamic, resource-limited healthcare settings because they can operate without heavy cryptographic operations. However, most remain rule-driven and lack adaptive intelligence.

BFT-IoMT [15] is a blockchain-enabled fuzzy trust management framework for large-scale IoMT deployments. It combines blockchain-based identity recording with fuzzified trust scoring using metrics such as communication consistency, residual energy, and packet reliability. The two-layer architecture (IoMT sensing layer and fog-based decision layer) supports distributed monitoring and tamper-resistant trust reporting. While BFT-IoMT improves Sybil detection accuracy and reduces single points of failure, it introduces significant computational and synchronization overhead due to blockchain operations and does not incorporate contextual (physiological or environmental) information. Furthermore, once nodes are revoked, the framework does not provide automated self-healing or reconfiguration, a critical limitation for WBAN where sensor availability has direct clinical implications.

TB2PA (Trust-Based Two-Phase Algorithm) [21] uses a hierarchical detection strategy: suspicious nodes are initially flagged based on communication path anomalies, then validated via aggregated direct and indirect trust values. This dual-stage model improves detection accuracy and reduces false positives by combining topology-based analysis with behavioral trust evaluation. However, trust metrics are updated at fixed intervals and rely on static rules. The algorithm assumes relatively stable topology and is thus less suitable for highly mobile WBANs where body movement continuously alters communication patterns.

LETM-IoT [22] proposes a more lightweight and multi-type Sybil detection framework, tracking behavioral instability in parent–child relationships, packet sequences, and identity swapping patterns. It classifies attacks into Type A, B, or C, demonstrating high true positive rates and lower memory consumption, making it more compatible with constrained medical devices. Yet,it lacks long-term behavioral memory, contextual awareness, and autonomous mitigation. Once a node is flagged as malicious, it is isolated, but the network is not actively repaired or re-optimized.

Across trust-based solutions, three recurring limitations emerge:

- Limited context awareness – Trust is primarily inferred from network behavior rather than combined physiological or environmental signals.
- Absence of AI-driven adaptation – Most frameworks use fixed rules or fuzzy logic instead of learning-based models capable of evolving with changing attack strategies.
- No self-healing capability – Networks detect Sybil nodes but do not autonomously reconfigure routes or restore services after isolation.

These limitations underscore the need for a context-aware, AI-enhanced, and self-healing trust framework tailored to WBAN-centric IoMT systems.

B. Context-Aware Approaches

Context-aware defenses extend beyond static identity or rule checking by incorporating dynamic network, behavioral, and sometimes physiological context into the detection process. This aligns well with the mobility, heterogeneity, and energy constraints of medical sensor networks. A first line of work focuses on network-context features such as routing patterns, RSSI variations, and signal behavior. SAPA-BAN [9] evaluates node identity and routing context to detect forged identities during cluster formation, offering lightweight protection in WBANs. Its effectiveness, however, depends on assumptions about recognizable malicious identity structures, limiting its scalability against stealthier adversaries.

The RSSI-LEA-AODV protocol [10] combines physical-layer RSSI measurements with encryption and routing adjustments to detect and avoid Sybil nodes. Once a node is identified as suspicious based on RSSI deviation, secure routing is re-established to bypass it. This layered approach strengthens protection against identity spoofing, but real-world RSSI fluctuations can cause detection uncertainty and false alarms.

Recent work extends context awareness with mobility modeling [23]. A collaborative edge-based detection model uses time-varying signal strength and learned mobility patterns (e.g., LSTM-based prediction) to identify unrealistic movement behaviors associated with Sybil identities. Offloading computation to edge nodes reduces sensor-side overhead, making it more suitable for constrained IoMT deployments. Nevertheless, performance is highly dependent on accurate motion models, and false positives remain a challenge.

A second line of research considers behavioral and role-based context, examining system roles, resource usage, and interaction patterns. SybilGuard [4], for instance, integrates lightweight authentication, access policy enforcement, and behavior-triggered verification. Rather than continuous monitoring, verification is activated upon policy violations, reducing energy consumption. While efficient, this selective strategy may overlook passive Sybil nodes and is vulnerable to single points of failure due to centralized cluster-head roles.

More advanced frameworks incorporate multi-modal context, combining network telemetry with biomedical sensor data [13], [23], [24]. Deep learning–based intrusion detection systems, for example, jointly analyze biometric signals and network behavior to identify anomalies. Although this improves detection accuracy, it raises privacy concerns and substantially increases computational demands, often exceeding what WBAN nodes can support.

In summary, context-aware solutions show substantial promise but exhibit common gaps: reliance on a single context dimension, difficulty balancing accuracy with energy efficiency, limited attention to privacy, and lack of integration with autonomous recovery mechanisms. Few models explicitly target WBAN-specific physiological mobility patterns, and none fully couple context-aware detection with self-healing mitigation.

C. Cryptographic and Blockchain-Oriented Approaches

Cryptographic techniques in WBAN security aim to guarantee data confidentiality, integrity, and authenticity, and to prevent device impersonation. Due to strict energy and processing constraints, most solutions employ simplified or lightweight schemes, often combined with non-cryptographic methods for Sybil detection.

In the SybilGuard framework [4], a custom lightweight encryption scheme is integrated with prime-order grouping to secure Smart Health Records (SHRs). The algorithm operates on 64-bit plaintext with 32-bit random keys in multiple rounds that include basic operations such as encoding, pair swapping, XOR, and one's complement. Prime-order group structures offer security comparable to 1024-bit composite groups while using a much smaller sequence group, substantially improving speed. Policy-hiding techniques further protect access policies. Although this design effectively safeguards data confidentiality for resource-limited devices, Sybil attack mitigation still relies on separate, non-cryptographic detection and attribute tracing modules.

RSSI-LEA-AODV [10] employs a hybrid cryptographic layer combining the classical Caesar Cipher Algorithm with a modern Lightweight Encryption Algorithm (LEA). Data undergo two encryption stages: an initial Caesar shift and subsequent LEA-based block encryption with 64-bit symmetric keys and key expansion for confusion and diffusion. While LEA is suitable for constrained devices, the reliance on Caesar cipher—cryptographically weak and primarily educational—illustrates the extreme trade-offs often made for performance. Importantly, Sybil identification is still driven by RSSI checks and energy-aware routing, while the cryptography primarily protects payload confidentiality.

BFT-IoMT [15] incorporates blockchain to provide tamper-resistant identity management. Device identities and trust metrics are stored on an immutable ledger, secured through cryptographic hashing. This decentralized architecture removes central authorities as single points of failure, thereby offering strong protection against identity manipulation and Sybil attacks. However, Blockchain synchronization and fog-layer computation introduce considerable overhead compared to lightweight local methods such as SybilGuard, making BFT-IoMT less suitable for ultra-constrained WBAN nodes.

In contrast, SAPA-BAN [9] explicitly follows a non-key-based design, foregoing encryption altogether. Although it claims to provide confidentiality and integrity through path re-selection and avoidance of malicious nodes, this contradicts standard cybersecurity principles, which regard encryption as fundamental for confidentiality. As a result, SAPA-BAN remains highly vulnerable to advanced sniffing and endpoint attacks.

Overall, cryptographic and blockchain approaches can significantly strengthen identity assurance and data protection but

often clash with WBAN resource budgets. Their overhead, and the tendency to focus more on payload security than identity-level manipulation, limits their standalone effectiveness for Sybil mitigation in body-area networks.

D. Machine Learning–Based Approaches

Machine learning (ML) has emerged as a promising direction for Sybil and intrusion detection in constrained wireless medical networks. ML-based defenses leverage pattern recognition and anomaly detection to identify attacks without exhaustively specifying rule sets. However, many existing ML frameworks are not yet optimized for real-time, energy-aware WBAN operation.

One representative work, "Detecting Sybil Attacks in Wireless Sensor Networks Using Machine Learning Algorithms" [25], applies supervised learning to classify multiple attack types—including Sybil, DDoS, R2L, Probe, and U2R—using the NSL-KDD dataset. The authors evaluate SVM, Random Forest, Logistic Regression, and Decision Tree models, with SVM achieving approximately 93% accuracy. While results are encouraging, the system focuses solely on detection, with no built-in prevention, mitigation, or self-recovery mechanisms. Moreover, NSL-KDD is a generic network dataset; it does not reflect WBAN or IoMT-specific traffic patterns, limiting direct applicability to medical environments.

A broader perspective is provided by "Cyber-Attack Detection in IoMT Using Machine Learning Techniques" [26], which reviews ML frameworks deployed at fog and cloud layers for healthcare security. The study emphasizes the advantages of distributed learning but also notes challenges such as heterogeneous sensor formats, scarcity of labeled medical datasets, and complexities in configuring ML models for real-time IDS. It calls for hybrid anomaly detection that combines network behavior with vital sign patterns and environmental context. Yet, the reviewed systems remain largely detection-oriented and lack autonomous mitigation and self-healing capabilities.

"Advancements in Sybil Attack Detection: A Comprehensive Survey of Machine Learning-Based Approaches in Wireless Sensor Networks" [27] analyzes supervised, semi-supervised, and deep learning methods, including k-NN, decision trees, SVM, cluster-based unsupervised learning, and deep reinforcement learning. Although ensemble and supervised models often exceed 95% accuracy in controlled setups, they typically require labeled data and considerable computational resources, which are impractical for miniature WBAN devices. The review emphasizes the need for adaptive, lightweight ML models tuned to constrained, mobile sensor platforms.

In a more healthcare-focused implementation, HealthGuard [24] proposes an IDS for smart healthcare that monitors multi-sensor physiological data (ECG, glucose, neural signals, etc.) and uses ANN, Random Forest, Decision Tree, and KNN to detect malicious behaviors, including false data injection and DoS. With average detection accuracy around 91%, HealthGuard highlights the benefits of integrating physiological context with ML-based detection. Nonetheless, it remains a passive system: it raises alerts but does not perform automated isolation, re-authentication, or self-reconfiguration.

Advanced frameworks such as IoT-AS Adaptive Security and deep learning–based IDS for IoMT [13] explore CNN–LSTM hybrids with attention mechanisms,achieving detection rates close to 99%.These solutions showcase the potential of deep learning for feature extraction, even on imbalanced datasets. However, they focus on general IoT security rather than WBAN-specific constraints, and their computational requirements are better suited to fog or cloud resources than to body-worn nodes.

Recent work in broader IoT security also presents adaptive, AI-driven self-healing mechanisms [28] . Some frameworks achieve high classification accuracy and dynamic automated response, significantly reducing recovery times by integrating anomaly detection, causal analysis, and automated remediation. Although promising conceptually, these systems are not yet specialized for Sybil identity spoofing or the strict latency and energy requirements of medical WBANs.

Across ML-based approaches, persistent limitations include:

- Emphasis on detection over mitigation or self-healing;
- Dependence on generic, non-medical datasets rather than WBAN-specific traffic and physiological signals;
- High computational overhead incompatible with wearable devices; and
- Limited integration of multi-modal context to distinguish genuine emergencies from adversarial behavior.

These limitations point to the need for lightweight, context-rich, and self-healing ML frameworks specifically designed for WBAN-based IoMT.

IV. Comparative Analysis

A. Comparative Evaluation of Existing Approaches

Existing Sybil attack defense mechanisms for IoMT/WBAN can be broadly grouped into trust-based, context-aware, cryptographic/blockchain-based, and machine learning (ML) approaches. Each has demonstrated utility in specific scenarios, yet none fully addresses the stringent requirements of WBAN-based healthcare networks.

A summarized comparison is provided in Table 1.

B. Critical Observations

- Trade-offs Between Security and Resource Constraints

Lightweight trust- or context-based methods reduce computational load but often sacrifice accuracy or multi-context integration. Cryptographic and blockchain mechanisms improve identity verification and data integrity but are often too heavy for low-power WBAN nodes, introducing delays and energy drain. ML-based approaches promise adaptability but require offloading or substantial local computation.

- Detection vs. Mitigation Gap

Most studies emphasize detection of Sybil nodes. Few systems incorporate self-healing mechanisms such as automatic route

TABLE I
SUMMARIZED COMPARISON OF SYBIL DEFENSE APPROACHES IN IOMT/WBAN.

Approach	Strengths	Limitations	Relevance to WBAN/IoMT
Trust-Based (e.gTB2PA, LETM-IoT, BFT-IoMT)	Lightweight, interpretable, can operate without heavy encryption; some frameworks (BFT-IoMT) provide decentralized trust via blockchain	Fixed rules or fuzzy logic, lack adaptive intelligence, limited contextual awareness, no autonomous recovery, may not scale with high mobility	Effective for low-complexity WBANs; struggles with highly mobile or multi-hop IoMT deployments
Context-Aware (e.g.,SAPA-BAN,RSSI-LEA-AODV, SybilGuard)	Integrates network, physical, and behavioral signals; lightweight implementations feasible; can use RSSI, mobility patterns for detection	Often single-context focused, sensitive to environmental fluctuations (RSSI), limited multi-modal fusion, lacks automated network healing	Highly relevant to WBANs where mobility and topology are constrained; detection alone may be insufficient
Cryptographic/Blockchain-Based (e.g.,SybilGuard encryption, BFT-IoMT)	Strong authentication, immutability, tamper-proof identity verification; prevents payload manipulation	High computational and energy overhead, blockchain latency, often focuses on payload security rather than real-time Sybil mitigation	Less suited for energy-constrained WBAN nodes; better suited for fog-level or multi-device IoMT setups
Machine Learning-Based (e.g., HealthGuard, CNN-LSTM IDS)	High detection accuracy; potential to integrate network, device, and physiological context; adaptive models	Resource-intensive, dependent on large labeled datasets, often offline, lacks autonomous mitigation/self-healing	Promising for hybrid fog/edge environments; direct deployment on wearable WBAN nodes is impractical without optimization

reconfiguration, node isolation, or real-time recovery of physiological data streams. This is a critical limitation in WBANs, where uninterrupted monitoring is essential for patient safety.

- Limited Multi-Modal Context Integration

Current approaches largely rely on a single dimension—network behavior, RSSI, trust score, or cryptographic verification. They rarely combine physiological signals, mobility patterns, and network telemetry, which is crucial for differentiating between legitimate sensor anomalies and adversarial Sybil attacks.

- Static or Predefined Models

Many trust-based or context-aware methods use fixed thresholds or static fuzzy logic, rendering them vulnerable to adaptive attackers who vary their Sybil patterns over time. Similarly, ML-based frameworks often lack online adaptation or incremental learning to cope with evolving attack strategies.

- Scalability and Mobility Limitations

WBAN networks may operate in highly dynamic environments (e.g., hospital wards, patient mobility). Existing frameworks do not adequately address distributed, mobile Sybil attacks (SA-II and SA-III), which can affect multiple sensors and gateways simultaneously, leading to larger-scale disruptions in IoMT settings.

V. RESEARCH GAP

Despite substantial progress in Sybil attack detection for IoMT and WBAN environments, current research still fails to deliver a lightweight, context-aware, and truly autonomous security framework suitable for the stringent constraints of medical WBAN deployments [3], [4], [22].

A. Lack of Integrated Multi-Layer Sybil Defense

Existing studies address Sybil attacks through isolated mechanisms such as cryptographic authentication [9], [10], trust scoring [15], [22], [31], RSSI-based identity checks [10], or machine learning classification [25], [27]. However, no unified architecture combines:

- Lightweight identity verification,
- Behavioral trust evaluation,
- Context-aware anomaly detection,
- Energy-efficient cryptographic support.

This fragmentation leaves exploitable gaps, enabling attackers to bypass individual layers of defense as highlighted in recent comprehensive reviews [12], [14]

B. Incomplete Coverage of Sybil Attack Types (SA-I, SA-II, SA-III)

Most WBAN security solutions consider only simplified, static identity forgeries [11], [32]. Very few address the full Sybil spectrum:

- **SA-I**: Local Static Sybil,
- **SA-II**: Distributed/Remote Sybil,
- **SA-III**: Mobile & Dynamic Sybil (most harmful to WBAN) [5].

The SA-III variant, which directly affects mobile and body-centric environments, is the most relevant to IoMT/WBAN but remains the least studied due to its complexity [23], [33].

C. Absence of Self-Healing and Autonomous Recovery

Current WBAN-focused defenses stop at detection [4], [17]. Typical responses include:

- Raising alerts,
- Isolating nodes,
- Updating blacklists.

No existing system provides automatic self-recovery capabilities such as:

- Dynamic re-routing of physiological data,
- Re-establishing secure communication paths,
- Re-validating legitimate sensors,
- Maintaining continuous monitoring without human intervention.

This gap is critical because medical environments cannot tolerate interruptions in monitoring. While general IoT self-healing concepts exist [28], [37]. they have not been adequately adapted for the specific routing and physiological constraints of WBANs.

D. Insufficient Multi-Modal Context Fusion

Most mechanisms rely on a single-context signal (e.g., trust metrics [21], RSSI [10], or traffic features). Very few fuse:

- Physiological signals (ECG, motion, heart rate) [24],
- Network-layer behavior [35],
- Device mobility patterns [23],
- Environmental context.

Without multi-modal fusion, systems struggle to distinguish between:

- Real clinical emergencies,
- Sensor faults,
- Intentional Sybil manipulation,

resulting in high false positives and unreliable detection [6].

E. Static Models Vulnerable to Adaptive Attackers

Trust-based, rule-based, and fuzzy-logic defenses often rely on fixed thresholds [15], [31]. Machine learning models are typically:

- Trained offline,
- Non-adaptive,
- Not updated during operation [26].

Consequently, adaptive attackers can change their behavior to evade these static systems [16].

F. Lack of Lightweight AI Optimized for WBAN Constraints

Deep learning IDS models proposed for general IoT are:

- Too computationally heavy,
- Designed for fog or cloud deployment,
- Not tailored to ultra-low-power wearable nodes.

A major gap remains in developing WBAN-appropriate, energy-aware AI capable of operating under:

- Millisecond-level latency,
- Tiny memory footprints,
- Strict battery limitations.

G. No Joint Optimization of Security, Energy, and Clinical QoS

Current studies treat security, energy efficiency, and data delivery latency as separate objectives [3], [36]. However, a practical WBAN security framework must jointly optimize all three. No existing work provides such holistic, clinically aligned optimization [37].

VI. Future Research Directions

Building on the identified research gaps and limitations in existing Sybil attack defenses for IoMT and WBAN environments [8], [29], several strategic future research directions emerge. These directions emphasize resilience, adaptability, clinical safety, and practical deployability in highly resource-constrained medical systems. The following subsections outline guided, actionable pathways that future researchers can pursue.

A. Toward a Unified Multi-Layer Security Architecture

Future work should focus on designing an integrated, multi-layered WBAN security architecture that incorporates:

- Lightweight cryptographic identity verification [18],
- Behavioral and trust-score–based monitoring [22],
- Context-aware mobility and network analysis [23],
- Adaptive machine learning–based intrusion detection [16].

Such integration would eliminate single-point security dependencies and close fragmentation gaps between existing isolated mechanisms. A unified architecture should also define clear inter-layer communication pathways, enabling cross-verification of sensor identity, mobility consistency checks, and anomaly-based behavioral validation. This direction encourages researchers to investigate how architectural layering can be optimized for ultra-low-power devices without introducing excessive latency [12].

B. Fully Autonomous Self-Healing WBAN Networks

Future WBAN designs must move beyond passive detection toward active, autonomous recovery [34]. Key research tasks include:

- Developing algorithms for real-time route reconstruction after Sybil node isolation [35], [38],
- Designing autonomous restoration mechanisms for disrupted physiological data streams,
- Creating dynamic reassignment protocols for cluster heads or gateway nodes,
- Enabling secure re-authentication and reintegration of falsely isolated legitimate nodes [28].

Researchers should explore proactive healing policies where WBANs predict failures using network risk indicators. Formal models, digital twins of human physiology, and hardware-based prototypes will be crucial for validating these healing mechanisms under real-world mobility, posture changes, and adversarial conditions.

C. Multi-Modal, Context-Aware Intelligence

A major future direction is the fusion of multiple contextual sources to improve Sybil detection accuracy. Research efforts should investigate how to jointly model:

- Network telemetry and traffic anomalies [17],
- Device behavior and mobility patterns [23],
- Physiological signals (ECG, PPG, HRV, temperature) [24],

- Environmental or activity-based context.

This multi-modal fusion should help systems differentiate:

- Clinical emergencies,
- Sensor faults or misplacements,
- Coordinated Sybil intrusions.

Future work should explore probabilistic fusion models, graph-based contextual reasoning, and privacy-preserving context extraction to ensure both safety and compliance with clinical regulations [2], [36].

D. Lightweight, Adaptive, and Explainable AI for WBAN Nodes

Given the stringent hardware constraints of WBAN nodes, research should prioritize the development of:

- TinyML techniques, model pruning, quantization, and on-node compression [8],
- Online and incremental learning frameworks capable of adapting to evolving Sybil behaviors [16],
- Adversarially robust ML models resilient to crafted manipulation attempts,
- Clinically interpretable Explainable AI (XAI) modules for decision transparency.

These directions encourage the design of ML models that support continuous learning with minimal energy impact, ensuring secure real-time inference without sending sensitive data to external servers [6].

E. Energy-Aware, Multi-Objective Optimization

Future WBAN security frameworks must jointly optimize:

- Sybil detection accuracy,
- Energy consumption,
- Communication overhead,
- End-to-end latency.

Researchers should explore multi-objective optimization algorithms (e.g., Pareto-based evolutionary methods) to balance clinical requirements with device limitations [37]. This includes modeling the trade-offs between detection precision and battery longevity, especially during long-term patient monitoring scenarios [3].

F. Expanded Attack Spectrum and Mobility-Aware Defenses

Sybil attack research must expand to consider increasingly sophisticated adversaries. Future studies should address:

- Combined attacks such as Sybil + wormhole, sinkhole, or jamming [5], [32],
- Mobility-induced vulnerabilities due to natural human movement [33],
- Full-spectrum Sybil variants: SA-I (localized static), SA-II (distributed static), SA-III (mobile/distributed).

A priority direction is the design of detection models that remain robust despite fluctuating RSSI values, posture changes, and signal distortions caused by body shadowing and limb movement. Mobility modeling, wearable-to-body propagation studies, and physics-informed detection algorithms can significantly advance this field [10], [23].

G. Standardized WBAN Datasets and Evaluation Platforms

Progress is currently limited by the lack of public datasets. Future research should develop:

- WBAN-specific datasets combining physiological signals and network traffic traces [**?**],
- Realistic simulations of SA-I, SA-II, and SA-III attack behaviors,
- Open-source benchmark platforms enabling standardized, reproducible evaluation.

Such datasets should reflect real patient motion, sensor misplacements, multi-hop routing, and interference conditions. Standardization will enable fair comparison and accelerate industry adoption.

H. Regulatory, Clinical, and Deployment-Aware Security Design

Security solutions for WBANs must integrate technical innovation with clinical practicality. Future work should examine:

- Compliance with medical device safety standards [39],
- Alignment with privacy regulations such as GDPR and HIPAA [2], [12],
- Consideration of real-world hospital workflows, emergency protocols, and interoperability requirements.

This direction encourages developing frameworks suitable for heterogeneous IoMT ecosystems, ensuring that security does not interfere with medical reliability, usability, or patient safety. Researchers should explore human-centered design, clinician-in-the-loop methods, and deployment testing in real medical environments [1], [36].

VII. Conclusion

This review underscores the growing complexity and urgency of securing Wireless Body Area Networks within the broader IoMT ecosystem. As healthcare systems increasingly rely on continuous monitoring through wearable and implantable sensors, failures in data confidentiality, integrity, or availability translate directly into risks for patient safety. Sybil attacks, by enabling one malicious device to impersonate many, represent a particularly damaging threat: they disrupt routing, drain node energy, corrupt clinical data flows, and erode trust in automated, life-critical applications.

Existing defense mechanisms—spanning trust-based, cryptographic, context-aware, and machine learning paradigms—provide important building blocks but remain individually insufficient for WBAN settings. Trust-based approaches are largely rule-driven and static; cryptographic schemes introduce overhead or focus primarily on payload rather than identity; context-aware methods are vulnerable to environmental variability and often narrow in scope; and ML-based systems frequently rely on non-medical datasets and offer detection without autonomous mitigation or recovery. A consistent limitation across all categories is the lack of integrated self-healing: networks can identify

malicious behavior but cannot automatically reorganize or preserve continuous, safe service after an attack.

These findings define a clear research opportunity: a lightweight, context-aware, AI-driven, self-healing security framework capable of operating within the strict energy, computation, and latency constraints of WBAN devices while offering real-time Sybil detection, adaptive decision-making, and automated mitigation. Such a framework must fuse behavioral, network, and physiological context to differentiate genuine clinical events, sensor faults, and adversarial manipulation, and must autonomously restore a safe and functional network state following attacks.

By mapping the limitations of current solutions and articulating the specific requirements of medical WBAN environments, this literature review establishes a strong scientific basis for the proposed context-aware self-healing Sybil defense framework and its contribution to the secure evolution of patient-centered IoMT systems.

References

[1] S. Razdan and S. Sharma, "Internet of Medical Things (IoMT): Overview, Emerging Technologies, and Case Studies," IETE Technical Review, vol. 39, no. 4, pp. 1-14, May 2021, doi: https://doi.org/10.1080/02564602.2021.1927863.

[2] M. Mushtaq, M. A. Shah, and A. Ghafoor, "THE INTERNET OF MEDICAL THINGS (IOMT): SECURITY THREATS AND ISSUES AFFECTING DIGITAL ECONOMY," IEEE Xplore, Jun. 01, 2021. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9650945 (accessed May 11, 2022).

[3] T. Saba, K. Haseeb, I. Ahmed, and A. Rehman, "Secure and energy-efficient framework using Internet of Medical Things for e-healthcare," Journal of Infection and Public Health, vol. 13, no. 10, pp. 1567-1575, Oct. 2020, doi: https://doi.org/10.1016/j.jiph.2020.06.027.

[4] J. Li and Z. Wang, "Sybil Attack Detection for Secure IoT-Based Smart Healthcare Environments," Journal of The Institution of Engineers (India): Series B, vol. 105, no. 6, pp. 1557-1569, Apr. 2024, doi: https://doi.org/10.1007/s40031-024-01060-9.

[5] Shameera A, "Baseline Evaluation of Wireless Body Area Network (WBAN) Routing Protocols in the Presence of Sybil Attacks," IOSR Journal of Engineering (IOSRJEN), vol. 15, no. 4, Apr. 2025. Available: https://www.iosrjen.org/Papers/vol15_issue4/15040104.pdf.

[6] Ayoub Si-Ahmed, Mohammed Ali Al-Garadi, and Narhimene Boustia, "Survey of Machine Learning based intrusion detection methods for Internet of Medical Things," Applied Soft Computing, vol. 140, pp. 110227-110227, Jun. 2023, doi: https://doi.org/10.1016/j.asoc.2023.110227.

[7] M. Salayma, A. Al-Dubai, I. Romdhani, and Y. Nasser, "Wireless Body Area Network (WBAN)," ACM Computing Surveys, vol. 50, no. 1, pp. 1-38, Apr. 2017, doi: https://doi.org/10.1145/3041956.

[8] Muntadher Alsabah et al., "A comprehensive review on key technologies toward smart healthcare systems based IoT: technical aspects, challenges and future directions," Artificial Intelligence Review, vol. 58, no. 11, Aug. 2025, doi: https://doi.org/10.1007/s10462-025-11342-3.

[9] Rohit Kumar Ahlawat, A. Malik, and A. Sadhu, "Sybil Attack Prevention Algorithm for Body Area Networks," Advances in intelligent systems and computing, pp. 125-134, Oct. 2017, doi: https://doi.org/10.1007/978-981-10-6747-1_15.

[10] A. S. S. Thuluva, M. S. Somanathan, R. Somula, S. Sennan, and D. Burgos, "Secure and efficient transmission of data based on Caesar Cipher Algorithm for Sybil attack in IoT," EURASIP Journal on Advances in Signal Processing, vol. 2021, no. 1, Jul. 2021, doi: https://doi.org/10.1186/s13634-021-00748-0.

[11] A. O. Bang and U. P. Rao, "A novel decentralized security architecture against sybil attack in RPL-based IoT networks: a focus on smart home use case," The Journal of Supercomputing, May 2021, doi: https://doi.org/10.1007/s11227-021-03816-2.

[12] A. Saxena and S. Mittal, "Internet of Medical Things (IoMT) Security and Privacy: A Survey of Recent Advances and Enabling Technologies," Proceedings of the 2022 Fourteenth International Conference on Contemporary Computing, Aug. 2022, doi: https://doi.org/10.1145/3549206.3549301.

[13] V. Ravi, T. D. Pham, and M. Alazab, "Deep Learning-Based Network Intrusion Detection System for Internet of Medical Things," IEEE Internet of Things Magazine, vol. 6, no. 2, pp. 50-54, Jun. 2023, doi: https://doi.org/10.1109/iotm.001.2300021.

[14] A. Ghubaish, T. Salman, M. Zolanvari, D. Unal, A. K. Al-Ali, and R. Jain, "Recent Advances in the Internet of Medical Things (IoMT) Systems Security," IEEE Internet of Things Journal, vol. 8, no. 11, pp. 1-1, 2020, doi: https://doi.org/10.1109/jiot.2020.3045653.

[15] S. E Ali, N. Tariq, F. Aslam Khan, M. Ashraf, W. Abdul, and K. Saleem, "BFT-IoMT: A Blockchain-Based Trust Mechanism to Mitigate Sybil Attack Using Fuzzy Logic in the Internet of Medical Things," MDPI, vol. 23, no. 9, Apr. 2023, doi: https://doi.org/10.3390/s23094265.

[16] Dr. S. Kuraku, "Adaptive Security Framework For Iot: Utilizing AI And ML To Counteract Evolving Cyber Threats," Educational Administration Theory and Practices, Apr. 2023, doi: https://doi.org/10.53555/kuey.v29i4.6496

[17] A. Naghib, F. S. Gharehchopogh, and A. Zamanifar, "A comprehensive and systematic literature review on intrusion detection systems in the internet of medical things: current status, challenges, and opportunities," Artificial Intelligence Review, vol. 58, no. 4, Jan. 2025, doi: https://doi.org/10.1007/s10462-024-11101-w.

[18] K. Grewal Dangi and S. P. Panda, "Challenges in Wireless Body Area Network-A Survey," IEEE, doi: https://doi.org/10.1109/ICROIT.2014.6798337.

[19] M. Yaghoubi, K. Ahmed, and Y. Miao, "Wireless Body Area Network (WBAN): A Survey on Architecture, Technologies, Energy Consumption, and Security Challenges," Journal of Sensor and Actuator Networks, vol. 11, no. 4, p. 67, Oct. 2022, doi: https://doi.org/10.3390/jsan11040067.

[20] M. Jafari and F. Adibnia, "Securing IoMT healthcare systems with federated learning and BigchainDB," Future Generation Computer Systems, vol. 165, p. 107609, Apr. 2025, doi: https://doi.org/10.1016/j.future.2024.107609.

[21] A. Shaji and N. S Nair, "A NOVEL TRUST BASED TWO PHASE ALGORITHM TO DETECT SYBIL ATTACK IN IoMT NETWORKS," 2023, doi:https://doi.org/10.1109/ICSCC59169.2023.10334946.

[22] J. Hassan Nisar, A. Sohail, A. Ismail Awad, and M. A. Zaka, "LETM-IoT: A lightweight and efficient trust mechanism for Sybil attacks in Internet of Things networks," Ad Hoc Networks, vol. 163, Oct. 2024, doi: https://doi.org/10.1016/j.adhoc.2024.103576.

[23] J. Yan et al., "A novel Sybil attack detection scheme in mobile IoT based on collaborate edge computing," EURASIP Journal on Wireless Communications and Networking, vol. 2023, no. 1, Mar. 2023, doi: https://doi.org/10.1186/s13638-023-02233-8.

[24] A. I. Newaz, A. K. Sikder, M. A. Rahman, and A. S. Uluagac, "HealthGuard: A Machine Learning-Based Security Framework for Smart Healthcare Systems," IEEE Xplore, Oct. 01, 2019. https://ieeexplore.ieee.org/document/8931716

[25] M. Mounica, R. Vijayasaraswathi, and R. Vasavi, "Detecting Sybil Attack In Wireless Sensor Networks Using Machine Learning Algorithms," IOP Conference Series: Materials Science and Engineering, vol. 1042, no. 1, p. 012029, Jan. 2021, doi: https://doi.org/10.1088/1757-899x/1042/1/012029.

[26] M. Narang, A. Jatain, and N. Punetha, "A study on Cyber-attack detection in IoMT using Machine Learning Techniques," SSRN Electronic Journal, 2023, doi: https://doi.org/10.2139/ssrn.4387775.

[27] M. Anita, J Jenefa, D Vinodha, and M. Lapina, "Advancements in Sybil Attack Detection: A Comprehensive Survey of Machine Learning-Based Approaches in Wireless Sensor Networks," Oct. 16, 2024. https://www.researchgate.net/publication/384948686

[28] J. Ibitoye and E. Fatanmi, "SELF-HEALING NETWORKS USING AI-DRIVEN ROOT CAUSE ANALYSIS FOR CYBER RECOVERY," International Journal of Engineering Technology Research & Management, vol. 6, no. 12, Dec. 2022, doi: https://doi.org/10.5281/zenodo.16793124.

[29] "Wireless Body Area Networks and Their Applications—A Review," IEEE, vol. 11, pp. 9202-9220, Jan. 2023, doi: https://doi.org/10.1109/ACCESS.2023.3239008.

[30] Marwa Salayma, Ahmed Al-Dubai, Imed Romdhani, and Youssef Nasser. 2017. Wireless body area network (WBAN): A survey

on reliability, fault tolerance, and technologies coexistence. ACM Comput. Surv. 50, 1, Article 3 (March 2017), 38 pages. DOI: http://dx.doi.org/10.1145/3041956

[31] A. Almogren, I. Mohiuddin, I. Ud Din, H. Almajed, and N. Guizani, "FTM-IoMT: Fuzzy-Based Trust Management for Preventing Sybil Attacks in the Internet of Medical Things," IEEE Internet of Things Journal , vol. 8, no. 6, pp. 4485–4497, Mar. 2021, doi: https://doi.org/10.1109/JIOT.2020.3027440.

[32] K. Zhang, X. Liang, R. Lu, and X. Shen, "Sybil Attacks and Their Defenses in the Internet of Things," IEEE Internet of Things Journal, vol. 1, no. 5, pp. 372-383, Oct. 2014, doi: https://doi.org/10.1109/jiot.2014.2344013.

[33] A. Bochem and B. Leiding, "Rechained: Sybil-Resistant Distributed Identities for the Internet of Things and Mobile Ad Hoc Networks," Sensors, vol. 21, no. 9, p. 3257, Jan. 2021, doi: https://doi.org/10.3390/s21093257.

[34] G. Amit, A. Shabtai, and Y. Elovici, "A Self-Healing Mechanism for Internet of Things Devices," IEEE Security Privacy, 2020, doi: https://doi.org/10.1109/msec.2020.3013207.

[35] D. Airehrour, J. A. Gutierrez, and S. K. Ray, "SecTrust-RPL: A secure trust-aware RPL routing protocol for Internet of Things," Future Generation Computer Systems, vol. 93, pp. 860-876, Apr. 2019, doi: https://doi.org/10.1016/j.future.2018.03.021.

[36] Inas Al Khatib, Abdulrahim Shamayleh, and M. Ndiaye, "Healthcare and the Internet of Medical Things: Applications, Trends, Key Challenges, and Proposed Resolutions," Informatics, vol. 11, no. 3, pp. 47-47, Jul. 2024, doi: https://doi.org/10.3390/informatics11030047.

[37] A. Kanneboina and G. Sundaram, "Improving security performance of Internet of Medical Things using hybrid metaheuristic model," Multimedia Tools and Applications, Apr. 2024, doi: https://doi.org/10.1007/s11042-024-19188-7

[38] H.-S. Kim, J. Ko, D. E. Culler, and J. Paek, "Challenging the IPv6 Routing Protocol for Low-Power and Lossy Networks (RPL): A Survey," IEEE Communications Surveys Tutorials, vol. 19, no. 4, pp. 2502-2525, 2017, doi: https://doi.org/10.1109/comst.2017.2751617.

[39] K. S. Kwak, S. Ullah, and N. Ullah, "An overview of IEEE 802.15.6 standard," IEEE Xplore, Nov. 01, 2010. https://ieeexplore.ieee.org/document/5702867 (accessed Mar. 12, 2022).

[40] N. Accettura, Luigi Alfredo Grieco, G. Boggia, and P. Camarda, "Performance analysis of the RPL Routing Protocol," Apr. 2011, doi: https://doi.org/10.1109/icmech.2011.5971218.

[41] F. Gara, L. B. Saad, Rahma Ben Ayed, and B. Tourancheau, "RPL protocol adapted for healthcare and medical applications," HAL (Le Centre pour la Communication Scientifique Directe), Aug. 2015, doi: https://doi.org/10.1109/iwcmc.2015.7289167.